Alternating Current

1. In an ac circuit, V and I are given by $V = 100\sin(100t)$ volts, $I = 100\sin\left(100t + \frac{\pi}{3}\right) mA.$ The power dissipated in circuit is:
   1) 10^4 watt
   2) 10 watt
   3) 2.5 watt
   4) 5 watt
2. Alternating current can not be measured by dc ammeter because
   1) ac cannot pass through dc ammeter
   2) Average value of complete cycle is zero
   3) ac is virtual
   4) ac changes its direction
3. An alternating current in a circuit is given by $I = 20\sin(100\pi t + 0.05\pi) A.$ The r.m.s value and the frequency of current respectively are.
   1) 10 A & 100 Hz
   2) 10 A & 50 Hz
   3) $10\sqrt{2}A \& 50\,Hz$
   4) $10\sqrt{2}A \& 100\,Hz$
4. An alternating current of rms value 10 A is passed through a 12Ω resistor. The maximum potential difference across the resistor is
   1) 20 V
   2) 90 V
   3) 169.68 V
   4) None of these
5. If $E = 100\sin(100t)$ volt and $I = 100\sin\left(100t + \frac{\pi}{3}\right) mA.$ are the instantaneous value of voltage and current, then the rms value of voltage and current are respectively.
   1) 70.7V, 70.7mA
   2) 70.7V, 70.7A
   3) 141.4V, 141.4mA
   4) 141.4V, 141.4A
6. 220 V, 50 Hz, ac is applied to a resistor. The instantaneous value of voltage is.
   1) $220\sqrt{2}\sin 100\pi t$
   2) $220\sin 100\pi t$
   3) $220\sqrt{2}\sin 50\pi t$
   4) $220\sin 50\pi t$
7. If a current I given by $I_0 \sin\left(\omega t - \frac{\pi}{2}\right)$ flows in an ac circuit across which an ac potential of $E = E_0 \sin \omega t$ has been applied, then the power consumption P in the circuit will be

1) $P = \frac{E_0 I_0}{\sqrt{2}}$

2) $P = \sqrt{2} E_0 I_0$

3) $P = \frac{E_0 I_0}{2}$

4) $P = 0$

8. If an alternating voltage is represented as $E = 141\sin(628t)$. Then the *rms* value of the voltage and the frequency are respectively

1) 141V, 628 Hz
2) 100V, 50 Hz
3) 100V, 100 Hz
4) 141V, 100Hz

9. An ac source is rated at 220V, 50 Hz. The time taken for voltage to change from its peak value to zero is

1) $50\,\text{sec}$
2) $0.02\,\text{sec}$
3) $5\,\text{sec}$
4) $5\times10^{-3}\,\text{sec}$

10. The ratio of peak value and r.m.s value of an alternating current is.

1) 1
2) $\frac{1}{2}$
3) $\sqrt{2}$
4) $1/\sqrt{2}$

11. The impedance of a certain a. c. circuit is 50 ohms. If the net resistance in the circuit is 25 ohms then the power factor of the circuit will be

1) Zero
2) 0.5
3) 1
4) 0.25

12. A series R-C circuit is connected to an alternating voltage source. Consider tow situation

(a) When capacitor is air filled

(b) When capacitor is mica filled current through resistor is I and voltage across capacitor is V then

1) $V_a > V_b$
2) $i_a = i_b$
3) $V_a = V_b$
4) $V_a < V_b$

13. A resistance R' draws power 'P' when connected to an AC source. If an inductance is now placed in series with the resistance, such that the impedance of the circuit becomes 'Z', the power drawn will be

1) $P\sqrt{\frac{R}{Z}}$

2) $P\left(\frac{R}{Z}\right)$

3) P

4) $P\left(\frac{R}{Z}\right)^2$

14. In A.C. circuit having only capacitor, the current

1) Lags behind the volage by $\pi/2$ in phase
2) Leads the voltage by $\pi/2$ in phase
3) Leads the voltage by π in phase
4) Lags behind the voltage by π in phase

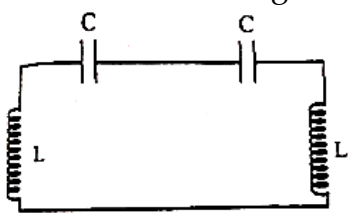
15. An arc lamp requires a direct current of 10 A at 80 V to function. If it is connected to a 220 V(rms), 50 Hz AC supply, the series inductor needed for it to work is close to

1) 0.08 H
2) 0.044 H
3) 0.065 H
4) 80

16. For which of the following reasons are LC oscillations not sustainable for long?

1) Every inductor has some resistance.
2) The effect of resistance is to introduce a damping effect on the charge & current in the circuit and the oscillations finally die away.
3) Even if the resistance is zero, the total energy of the system is radiated away from the system in the form of electromagnetic radiation
4) All of the above

17. The natural frequency of the circuit shown in the figure is:



1) $\frac{1}{2\pi\sqrt{LC}}$

2) $\frac{1}{\pi\sqrt{LC}}$

3) $\frac{2}{\pi\sqrt{LC}}$

4) None of these

18. In LC oscillation, what is the current in the circuit when the total energy is stored in the form of magnetic energy (q_o is the maximum charge stored by the capacitor)

1) Zero

2) $\frac{q_0}{\sqrt{LC}}$

3) $\frac{q_0}{LC}$

4) $q_0\sqrt{LC}$

19. An LC circuit contains an inductor L=25mh) and a capacitor (C = 25 mF) with an initial charge of Q_o. At what time will the circuit have an equal amount of electrical and magnetic energy?

1) $\frac{\pi}{160}s$

2) $\frac{3\pi}{160}s$

3) $\frac{5\pi}{160}s$

4) All of these

20. The transformer voltage induced in the secondary coil of a transformer is mainly due to

1) a varying magnetic field
2) a varying electric field
3) the vibrations of the primary coil
4) the iron core of the transformer

21. Power factor in LC oscillations is

1) Zero
2) 1
3) >1
4) <1

22. A step-down transformer connected to an ac mains supply of 220 V is made to operate at 11V, 44 W lamp. Ignoring power losses in the transformer, what is the current in the primary circuit?

1) 2 A
2) 4 A
3) 0.2 A
4) 0.4 A

23. A step-down transformer is used to reduce the main supply of 220 V to 10 V. If the primary draws 5 A and secondary draws 88 A current, calculate the efficiency of the transformer.

1) 60%
2) 70%
3) 80%
4) 90%

24. The natural frequency of a L - C circuit is equal to

1) $\frac{1}{2\pi}\sqrt{LC}$

2) $\frac{1}{2\pi\sqrt{LC}}$

3) $\frac{1}{2\pi}\sqrt{\frac{L}{C}}$

4) $\frac{1}{2\pi}\sqrt{\frac{C}{L}}$

25. A charged $30\mu F$ capacitor is connected to a 27 mH inductor. The angular frequency of free oscillations of the circuit is

1) $1.1\times10^{3}\ rad\ s^{-1}$
2) $4.1\times10^{3}\ rad\ s^{-1}$
3) $2.1\times10^{3}\ rad\ s^{-1}$
4) None of these

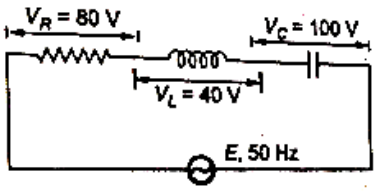
26. Power delivered by the source of the circuit becomes maximum, when

1) $\omega L=\omega C$
2) $\omega L=\frac{1}{\omega C}$
3) $\omega L=-\left(\frac{1}{\omega C}\right)^{2}$
4) $\omega L=\sqrt{\omega C}$

27. A resistor 30Ω, inductor of reactance 10Ω and capacitor of reactance 10Ω are connected in series to an a.c. voltage source $e=300\sqrt{2}\sin(\omega t)$. The current in the circuit is

1) $10\sqrt{2}A$
2) 10 A
3) $30\sqrt{11}\,A$
4) $\frac{30}{\sqrt{11}}A$

28. The value of alternating emf E in the given circuit will be.



1) 100
2) 20 V
3) 220 V
4) 140 V

29. The time constant of a L – R circuit is.

1) R/L
2) L/R
3) LR
4) $1/LR$

30. The potential differences across the resistance, capacitance and inductance are 80 V, 40 Vand 100 V respectively in an L-C-R circuit. The power factor of this circuit is.

1) 1.0
2) 0.4
3) 0.5

4) 0.8

Answer Key

(1)	3	**(16)**	4
(2)	2	**(17)**	1
(3)	3	**(18)**	1
(4)	3	**(19)**	1
(5)	1	**(20)**	1
(6)	1	**(21)**	2
(7)	4	**(22)**	3
(8)	3	**(23)**	3
(9)	4	**(24)**	2
(10)	3	**(25)**	1
(11)	2	**(26)**	2
(12)	1	**(27)**	2
(13)	4	**(28)**	1
(14)	2	**(29)**	2
(15)	3	**(30)**	4